2020-Summer Paper 1 – Basic Sciences in Urology

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Q1. Describe clinical implication of AIDS on the Genito-Urinary Tract?

Type: LAQ (25 marks)

Introduction

- AIDS (acquired immunodeficiency syndrome) is the late stage of HIV (human immunodeficiency virus) infection. HIV is a retrovirus that destroys CD4 T lymphocytes. It spreads by blood, semen, vaginal fluid and breast milk.

Stage	Features
1. Acute	2–4 weeks after infection; flu-like illness; high viral load; antibody tests may be negative
2. Clinical latency	Years; usually no symptoms
3. AIDS	**CD4 <200 cells/mm³** or an AIDS-defining condition

- **Concept:** genitourinary (GU) disease follows the falling CD4 count. Organisms become opportunistic, infections become chronic and the GU tract becomes a reservoir. With HAART (highly active antiretroviral therapy), the burden shifts to drug toxicity and diseases of ageing.

Diagnosis (urologist's role)

- Test anyone with, or at risk of, an STD (sexually transmitted disease), and every TB (tuberculosis) patient.
- Antigen–antibody immunoassay → antibody differentiation assay; if discordant, HIV RNA (acute infection).

Manifestations by site

Site	Manifestations
Genital skin and STDs	Ulcerative STDs (herpes, syphilis, chancroid) increase HIV acquisition and transmission; HSV-2 (herpes simplex virus 2) in 50–90%. Warts (HPV (human papillomavirus) 6, 11) in about 20%, may be giant; molluscum 5–18%. Perineal cellulitis (cover MRSA (methicillin-resistant *Staphylococcus aureus*)); Fournier gangrene commoner

Site	Manifestations
UTI (urinary tract infection)	About fivefold commoner. Bacteriuria about 30% at CD4 <200, 11% at 200–500, none above 500. *E. coli* and other gram-negatives; *Pseudomonas* up to 30%. *Salmonella* UTI needs lifelong prophylaxis. Culture before treatment
Opportunistic UTI	No response, or CD4 <500: look for fungi (*Candida*, *Cryptococcus*, *Histoplasma*), TB, CMV (cytomegalovirus), adenovirus
Kidney	Opportunistic pyelonephritis (*Pneumocystis*, *Candida*, mycobacteria). AKI (acute kidney injury) from sepsis, dehydration, nephrotoxins (tenofovir, indinavir, amphotericin). Obstruction by stones, lymphoma, fungal balls
GU TB	HIV raises active TB 15–20-fold. Disease is disseminated and bilateral, with **less fibrosis, fewer strictures and fewer contracted bladders**, because fibrosis needs an intact immune response. Mortality is higher. Same 6-month regimen; rifampicin lowers protease inhibitor levels
Prostate	Bacterial, TB and fungal prostatitis; prostatic abscess and urosepsis; relapse up to 70%. Cryptococcal prostatitis is a reservoir for relapsing meningitis, so give long-term antifungals
Testis and epididymis	Recurrent, chronic epididymo-orchitis (including TB). Testis harbours opportunists. Testicular atrophy and hypogonadism
Voiding	LUTS (lower urinary tract symptoms) worsen with disease stage. Retention may herald seroconversion. Neurogenic causes: HIV encephalopathy, toxoplasmosis, **CMV polyradiculopathy** (bladder and bowel dysfunction with sciatic pain: lumbar puncture, treat early), neuropathy
Sexual function	ED (erectile dysfunction) in about half of men under 50; protease inhibitors worsen it. **Protease inhibitors raise sildenafil levels 3–10-fold: start low.** Low libido, ejaculatory disorders

HIVAN (HIV-associated nephropathy)

- Nephrotic-range proteinuria and rising creatinine, often without oedema or hypertension. Typically CD4 <200 and viral load >1000 copies/mL; about 12 times commoner in Black patients.
- Biopsy: **collapsing FSGS (focal segmental glomerulosclerosis)** with tubular microcysts.
- Treatment: HAART plus ACE (angiotensin-converting enzyme) inhibitor. Without it, dialysis follows within about 10 months.

Stones

- **Indinavir** stones in up to 22% of treated patients. Indinavir precipitates in alkaline urine. Stones are radiolucent, and pure stones may be invisible even on CT (computed tomography).
- Stop the drug and hydrate: succeeds in up to 70%. Newer protease inhibitors rarely cause stones.

Malignancy

AIDS-defining	Non-AIDS-defining
Kaposi sarcoma (HHV-8 (human herpesvirus 8)), non-Hodgkin lymphoma, cervical cancer	Penile, anal, testicular, prostate cancers; Hodgkin lymphoma
Falls with HAART	Now about 70% of cancers in treated HIV

- **Kaposi sarcoma:** genital in about 3% of AIDS; glans lesions can block the meatus. Treat with ART (antiretroviral therapy) first. Local excision, laser or intralesional vinblastine; radiotherapy 15–30 Gy; systemic chemotherapy if disseminated.

- **Penile cancer:** HIV raises HPV infection; about fourfold risk of death.

Implications for urological practice

- **Universal precautions** for every patient: double gloves, no hand-to-hand passing of sharps.
- Occupational risk: percutaneous about 0.3%, mucous membrane about 0.09%.
- **PEP (post-exposure prophylaxis)** within 72 hours, for 28 days: tenofovir + emtricitabine plus raltegravir or dolutegravir. HIV test at 6 weeks, 3 and 6 months.
- CD4 <200: TMP-SMX (trimethoprim-sulfamethoxazole) *Pneumocystis* prophylaxis also lowers UTI.
- **Prevention:** male circumcision cuts heterosexual acquisition by 50–60% (three African RCTs (randomised controlled trials)); consistent condom use cuts risk about 80%; PrEP (pre-exposure prophylaxis).

Also know: IRIS (immune reconstitution inflammatory syndrome) on starting ART; serodiscordant couples: suppressed viral load with sperm washing and assisted reproduction; transplantation is possible with controlled HIV.

Diagram to draw: male GU tract outline labelled with the lesion at each site.

> **Key points to remember:**
> - AIDS = CD4 <200; UTI risk rises as CD4 falls.
> - HIV-TB: disseminated, less fibrosis, fewer strictures.
> - Indinavir stones: radiolucent, may be CT-invisible; stop drug and hydrate.
> - PEP within 72 hours for 28 days; circumcision cuts acquisition 50–60%.

Q2. Describe embryological development of VU Junction and outline management of primary VUR?

Type: LAQ (25 marks)

Part A: Embryology of the VUJ (vesicoureteric junction)

1. **Ureteric bud (weeks 4–6)** arises from the caudal mesonephric (Wolffian) duct: ureter and duct form a "Y", the common nephric duct is the stem.
2. **Reciprocal induction:** the bud enters the metanephric blastema, branches into the collecting system and induces nephrons.
3. **Weeks 5–6:** the urorectal septum divides the cloaca into urogenital sinus (UGS) and anorectal canal.
4. **Absorption:** the common nephric duct is taken into the UGS (current view: removed by **apoptosis**), so ureter and duct open separately.
5. **Rotation (Mackie and Stephens):** the ureteric orifice moves **cranially and laterally** to the bladder base; the duct stays caudal as the ejaculatory duct (Gartner duct in females).
6. **Trigone:** classically ureteral; lineage tracing shows mostly bladder muscle. **Chwalla membrane** over the orifice ruptures; persistence explains ureterocele.
7. **Maturation:** ureteric muscle joins the detrusor as **Waldeyer's sheath**; the tunnel lengthens with growth, so VUR (vesicoureteral reflux) resolves with age.

Bud origin	Orifice	Result
Normal	Trigone corner	Tunnel length to diameter about **5:1** (Paquin); refluxing junctions about 1.4:1
Too low (early contact)	High, **lateral**	Short tunnel → **primary VUR**; abnormal bud–blastema contact → dysplasia
Too high (late contact)	Caudal, **medial**	**Ectopic ureter** (bladder neck, urethra, Wolffian structures), often obstructed
Two buds	Duplex	**Weigert–Meyer rule:** upper pole caudal-medial (ectopia, ureterocele); lower pole lateral (reflux)

- **Antireflux mechanism:** passive flap valve, ureteric peristalsis, low bladder pressure.
- **Concept:** bud position fixes both orifice and kidney, so high-grade VUR and dysplasia coexist.

Part B: Management of primary VUR

- **Primary:** congenitally short tunnel with a normal bladder. **Secondary:** high bladder pressure (PUV (posterior urethral valves), neurogenic bladder, BBD (bladder and bowel dysfunction)).
- In about 30% of children with UTI (urinary tract infection) and about 30% of siblings.
- **Infection, not reflux, injures the kidney.** The aim is to prevent febrile UTI and scarring while VUR resolves.

International grading (VCUG, voiding cystourethrogram)

Grade	Findings
I	Ureter only, not dilated
II	Up to pelvis and calyces; no dilatation
III	Mild–moderate dilatation; minimal forniceal blunting
IV	Moderate dilatation and tortuosity
V	Gross dilatation and tortuosity; **papillary impressions lost**

Evaluation

- Urine culture; BP (blood pressure), creatinine, urine protein; BBD history.
- **RBUS (renal and bladder ultrasound)**; VCUG (gold standard); nuclear cystogram for follow-up; **DMSA (dimercaptosuccinic acid) scan** for scars.

Natural history: grade II resolves in 63–85%; grade III about 50%; grades IV–V and bilateral III rarely. Younger age resolves better.

Risk stratification: high risk = female, under 1 year, bilateral, grade III–V, BBD, scars, multiple febrile UTIs.

Medical management

- **Treat BBD first:** timed voiding, constipation treatment, anticholinergics.
- **CAP (continuous antibiotic prophylaxis):** one low dose at night. Trimethoprim or amoxicillin under 2 months; TMP-SMX (trimethoprim-sulfamethoxazole) or nitrofurantoin over 2 months.
- Boys with phimosis: circumcision or topical steroid (about 4 circumcisions prevent one UTI).
- **RIVUR (Randomized Intervention for Children with Vesicoureteral Reflux, 2014):** recurrent UTI fell from 23.6% to 12.9% on TMP-SMX; **scarring unchanged**.
- **Swedish Reflux Trial** (grade III–IV): febrile UTI in girls 19% on CAP, 23% after endoscopy, 57% on surveillance; no difference in boys.
- IRSC (International Reflux Study in Children), Birmingham: surgery equalled medical care for scarring.
- Low-risk children: observation without CAP.

Indications for correction: breakthrough febrile UTI on CAP; new scars or falling function; persistent high grade; non-adherence; parental choice.

Endoscopic injection: Dx/HA (dextranomer/hyaluronic acid) by STING (subureteric Teflon injection) or HIT (hydrodistension implantation technique); about 90% for low grade after one injection (see 2020-Summer Paper 4, Q3c).

Ureteric reimplantation: open success **95.9%** (Elder 1997).

Technique	Tunnel	Note
Cohen (intravesical)	Cross-trigonal	Commonest; over 98%; later ureteroscopy difficult
Politano–Leadbetter	New hiatus above old	Long tunnel; blind hiatus risks bowel
Glenn–Anderson	Advanced towards bladder neck	Least kinking; short tunnel
Lich–Gregoir (extravesical)	Detrusorotomy	No haematuria or spasm; **retention up to 10% after bilateral**
RALUR-EV (robotic extravesical)	Y-shaped detrusorotomy	Less pain; approaching open results

- Ultrasound at 6–12 weeks. Contralateral VUR in about 1 in 10, usually resolves.
- **Guidelines:** AUA (American Urological Association) 2010/2017 and EAU/ESPU (European Association of Urology/European Society for Paediatric Urology): treat BBD, CAP for high risk, correct after breakthrough infection.
- **Follow-up:** BP, growth; scarred women risk pre-eclampsia.

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Treat BBD -> low risk: observe | high risk: CAP
Breakthrough UTI or new scar -> injection or reimplant
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Diagram to draw: "Y" of ureteric bud and mesonephric duct with orifice migration; grades I–V.

Key points to remember:

- Low (early) bud → lateral orifice → VUR plus dysplasia; high (late) bud → ectopia.
- Tunnel length to diameter 5:1 (Paquin).
- RIVUR: CAP halves recurrent UTI, not scarring.
- Open reimplantation 95.9%; Cohen over 98%.

Q3a. Berger's Disease.

Type: SAQ (10 marks)

Berger's disease is IgA (immunoglobulin A) nephropathy (IgAN).

Definition

- **Commonest primary glomerulonephritis (GN) worldwide**, in children too. Dominant **mesangial IgA deposits** on immunofluorescence.
- Commonest cause of **glomerular haematuria** (about one-third). Second–third decades; male predominance.
- **IgA vasculitis** (Henoch–Schönlein purpura) is the systemic form.

Pathogenesis

- "Multi-hit" model: abnormal (galactose-deficient) IgA1 → autoantibodies → immune complexes → mesangial deposition, complement activation, proliferation. Mucosal (respiratory) infection triggers flares.

Clinical features

- **Recurrent painless macroscopic haematuria during or within days of an upper respiratory infection** ("synpharyngitic"), or after exercise. Post-streptococcal GN follows infection by weeks.
- Persistent microscopic haematuria, proteinuria, hypertension; occasional loin pain; rarely nephrotic syndrome or RPGN (rapidly progressive GN).

Evaluation (urologist's role)

- **Dysmorphic RBCs (red blood cells), RBC casts, proteinuria** → glomerular source. **Normal C3 and C4.**
- Without casts, complete the urological evaluation (imaging ± cystoscopy) to exclude a urinary tract cause.
- **Renal biopsy** confirms. Once diagnosed, repeat haematuria work-ups are not needed.

Oxford MEST-C score (biopsy)

Item	Grades
M: mesangial hypercellularity	M0, M1
E: endocapillary hypercellularity	E0, E1
S: segmental sclerosis	S0, S1
T: tubular atrophy/interstitial fibrosis	T0 ≤25%, T1 26–50%, T2 >50%
C: crescents	C0 none, C1 <25%, C2 ≥25% of glomeruli

Prognosis

- Often benign, but **20–30% reach ESKD (end-stage kidney disease)**.
- Poor: persistent proteinuria, hypertension, raised creatinine, older onset, high MEST-C.

Management

- No cure. **Supportive:** ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker), BP (blood pressure) control, low salt, stop smoking.
- **Newer drugs:** SGLT2 (sodium–glucose cotransporter 2) inhibitors; sparsentan (dual endothelin–angiotensin receptor blocker); targeted-release budesonide acting on gut mucosal lymphoid tissue.

- Persistent proteinuria above about 0.75–1 g/day despite supportive care: consider a 6-month steroid course (KDIGO (Kidney Disease: Improving Global Outcomes) 2021); weigh infection risk.
- Crescentic RPGN: steroids plus cyclophosphamide.
- **Transplant:** recurs histologically often, clinically rarely.

Diagram to draw: glomerulus with mesangial IgA deposits.

> **Key points to remember:**
> - Commonest primary GN; haematuria with upper respiratory infection.
> - Normal complement; biopsy shows mesangial IgA.
> - Oxford MEST-C; 20–30% ESKD.
> - ACE inhibitor or ARB is the mainstay.

Q3b. Describe instrumentation for development of extraperitoneal space in laparoscopy?

Type: SAQ (10 marks)

Introduction

- Extraperitoneal laparoscopy works outside the peritoneum: the **flank retroperitoneum** (kidney, adrenal, ureter) or the **pelvic prevesical (Retzius) space** (extraperitoneal prostatectomy, lymphadenectomy).
- There is no natural cavity, so a working space must be **dissected before insufflation**.

Instruments

Instrument	Description and use
Open (Hasson) access set	Scalpel, S-retractors, artery forceps
Index finger	Blunt first pocket; feels psoas behind, Gerota's fascia in front
Commercial balloon dissector	Transparent silicone balloon on a stiff hollow shaft; inflated with room air by bulb (maximum 800 mL, 40 pumps). Oblong for the retroperitoneum, round for the pelvis. Laparoscope inside confirms position
Gaur balloon (1992)	Size 7 glove (or glove fingers, condom) tied with 0 silk on an 8-Fr red rubber catheter; filled with saline (600–800 mL). Cheap, but no rigid shaft or view; latex can rupture into fragments
Optical (visual) obturator trocar with 0° scope	Enters under vision; yellow retroperitoneal fat confirms the plane; scope sweeps the space
Blunt-tip balloon trocar	Internal balloon plus external foam cuff gives an airtight seal; prevents gas leak, emphysema and hypercarbia

Technique

Retroperitoneal

1. Full flank, table flexed; 1.5–2.5-cm incision below the 12th rib tip.
2. Split muscles; open the lumbodorsal fascia; finger dissection between psoas and Gerota's fascia.
3. Balloon inflated cephalad then caudad, in front of psoas, behind Gerota's fascia.
4. Blunt-tip balloon trocar; insufflate; secondary ports under vision.

Pelvic extraperitoneal

- Infraumbilical incision → open anterior rectus sheath, separate recti → finger tunnels behind the rectus to the pubis → pierce transversalis fascia → balloon midline, then each side.

Pitfalls

- Balloon in muscle → flank hernia. Too anterior → peritoneal tear or colon injury; too posterior → psoas or quadratus bleeding.
- Peritoneal tear shrinks the space; convert if needed. Extraperitoneal CO_2 needs more ventilation.

Diagram to draw: flank section with balloon between psoas and Gerota's fascia.

Key points to remember:

- Space is created first: finger, then balloon.
- Gaur balloon: glove on red rubber catheter.
- Balloon trocar seals and prevents emphysema.
- Balloon in muscle causes hernia.

Q3c. Lichen sclerosis.

Type: SAQ (10 marks)

Taken as lichen sclerosus (LS), formerly balanitis xerotica obliterans (BXO).

Definition

- Chronic, progressive, **scarring inflammatory disease of genital skin**; possibly premalignant.

Epidemiology

- 6–10 times commoner in females (prepubertal and menopausal).
- Men: peak 30–50 years in one chapter, older men in another; also children. Commonly found at circumcision for phimosis after infancy. 15–20% have extragenital disease.

Aetiology

- Autoimmune (thyroid disease, vitiligo, alopecia); familial clustering.
- **Chronic occlusion by urine** under the foreskin (dribbling, obesity) definitely increases risk.
- Infection and trauma proposed, unproven.

Clinical features

- White, atrophic, sclerotic plaques; purpura, fissures; itch, pain, painful erection, poor stream.
- Sequence: prepuce → **phimosis** → glans → **meatal stenosis** (sentinel site) → fossa navicularis and penile urethral stricture.
- **Spares the proximal (bulbar and posterior) urethra.**
- Buried penis from loss of shaft skin.
- Mechanism of stricture: voiding against meatal stenosis forces urine into the glands of Littre → inflammation → spongiofibrosis.

Evaluation

- Clinical diagnosis. **Biopsy** if doubtful or suspicious: hyperkeratosis, epidermal atrophy, basal vacuolation, dermal oedema with homogenised collagen, band-like lymphocytic infiltrate.
- Uroflowmetry; **RUG (retrograde urethrogram) before meatal surgery**, because a proximal stricture often coexists.

Management

Situation	Treatment
Skin disease	**Clobetasol propionate 0.05%** daily for 1–3 months, then taper to twice-weekly maintenance; calcineurin inhibitors second line; retinoids for refractory disease
Phimosis	**Circumcision**, usually curative
Meatal stenosis	Dilatation with clobetasol; extended meatotomy; BMG (buccal mucosal graft) meatoplasty
Anterior stricture	**Oral mucosal graft, never genital skin**; staged or single stage
Panurethral or unfit	**Perineal urethrostomy** (easy because the proximal urethra is spared); self-dilatation

- Recurrence 40–50% after medical treatment; restricture higher than in other aetiologies.

Malignancy

- Penile SCC (squamous cell carcinoma) in **2.3–9%** on long follow-up; HPV (human papillomavirus)-independent type.
- **Lifelong follow-up**; biopsy any non-healing ulcer or nodule.

Diagram to draw: penis with LS at prepuce, glans, meatus and fossa, proximal urethra spared.

> **Key points to remember:**
>
> - Genital skin disease; meatus is the sentinel site.
> - Spares proximal urethra; perineal urethrostomy easy.
> - Clobetasol first; oral mucosa, never skin, for repair.
> - SCC in 2.3–9%: lifelong follow-up.

Q3d. Renal allograft rejection.

Type: SAQ (10 marks)

Definition

- Recipient immune attack on donor (allo) antigens: **HLA (human leucocyte antigen)** class I (A, B, C) and II (DR, DQ, DP), ABO and non-HLA antigens.
- Acute rejection has fallen from over 60% (pre-ciclosporin) to **below 10%**. **Chronic antibody injury** now limits graft survival.

Mechanism

- **T-cell**: direct (donor antigen-presenting cells) and indirect pathways → cytotoxic T cells → tubulitis, arteritis.
- **Antibody**: DSA (donor-specific antibody) binds endothelium → complement, leaving **C4d** → microvascular inflammation.

Types

Type	Timing	Features
Hyperacute	Minutes–hours	Preformed ABO/HLA antibody; thrombosis; graft turns blue. **Prevented by crossmatch**
Accelerated	2–5 days	Memory response in sensitised recipient
Acute TCMR (T-cell-mediated rejection)	Mostly first 6 months	Interstitial infiltrate, tubulitis, arteritis
Acute AMR (antibody-mediated rejection)	Any time	DSA, C4d, capillaritis
Chronic active AMR	Months–years	Transplant glomerulopathy, arterial intimal fibrosis; main cause of late loss
Subclinical	—	Histological rejection, stable function

- **Risk factors:** HLA mismatch, DSA, ABO incompatibility, long ischaemia and DGF (delayed graft function), **non-adherence**, retransplant, previous rejection.

Diagnosis

- Rising creatinine, oliguria; fever and graft tenderness now rare.
- **Exclude:** obstruction, urine leak, lymphocele, renal artery stenosis or thrombosis (ultrasound with Doppler), CNI (calcineurin inhibitor) toxicity, BK virus nephropathy, recurrent disease.
- **Biopsy is the gold standard (Banff):**

Banff category	Criteria
Borderline	Foci of tubulitis with mild inflammation
TCMR IA/IB	Interstitial inflammation >25% with moderate/severe tubulitis
TCMR IIA/IIB	Mild-moderate/severe intimal arteritis
TCMR III	Transmural arteritis or fibrinoid necrosis
Active AMR	Tissue injury + antibody–endothelium interaction (C4d) + circulating DSA

- **Monitoring:** de novo DSA predicts loss; dd-cfDNA (donor-derived cell-free DNA) has negative predictive value >90%; gene-expression panels.

Treatment

- **TCMR:** IV (intravenous) pulse methylprednisolone; **ATG (antithymocyte globulin)** for Banff II–III or steroid-resistant.
- **AMR:** not standardised: steroids, **plasmapheresis, IVIG (intravenous immunoglobulin)**, rituximab; eculizumab in selected patients; anti-IL-6 (interleukin-6) for chronic active AMR.
- Optimise maintenance (tacrolimus, mycophenolate, steroids); address adherence.

Diagram to draw: Banff biopsy features: tubulitis, arteritis, peritubular capillary C4d.

> **Key points to remember:**
>
> - Hyperacute rejection prevented by crossmatch.
> - Banff AMR = injury + C4d + DSA.
> - TCMR: pulse steroids, ATG; AMR: plasmapheresis, IVIG.
> - Chronic active AMR is the main cause of late loss.

Q3e. Leak point pressures-clinical implication.

Type: SAQ (10 marks)

Definition

- **LPP (leak point pressure):** pressure at which leakage is first seen, spontaneous or provoked. Two independent types:

Feature	ALPP (abdominal LPP), incl. VLPP (Valsalva LPP) and CLPP (cough LPP)	DLPP (detrusor LPP)
Definition	Total Pves (vesical pressure) at leak during raised abdominal pressure, **no detrusor contraction**	Lowest Pdet (detrusor pressure) at leak, **without contraction or straining**
Measures	Sphincter strength	Bladder pressure against outlet resistance
Bad result	**Low** (weak sphincter)	**High** (danger to upper tracts)
Used in	SUI (stress urinary incontinence)	Neurogenic or low-compliance bladder

VLPP: technique

1. Urethral and rectal catheters; standing; fill to **150 mL**; graded Valsalva; record **total Pves** at visible leak.
2. No leak: repeat every 50 mL; at capacity remove the urethral catheter and read the rectal line.

VLPP: clinical implications

VLPP (cmH_2O)	Inference (McGuire 1993)
<60	**ISD (intrinsic sphincter deficiency)**
60–90	Equivocal
≥90	**Urethral hypermobility**, little ISD

- Low VLPP (ISD) traditionally favours a sling or bulking agent over colposuspension.
- **Limitations:** VLPP < CLPP; falls with larger catheters and volumes; invalid with DO (detrusor overactivity); poor predictor of surgical outcome.
- **Guidelines:** EAU (European Association of Urology): do not use LPP to grade severity. AUA (American Urological Association) 2023 SUI guideline: urodynamics optional in the index patient.

DLPP: clinical implications

- **McGuire (1981):** DLPP **>40 cmH_2O** in myelodysplastic children → hydronephrosis or VUR (vesicoureteral reflux) in **85%**.
- A strong outlet keeps the patient dry but transmits pressure to the kidneys; a weak outlet "pops off" safely.
- **Hostile bladder** (≥40 cmH_2O) → CIC (clean intermittent catheterisation) plus antimuscarinic; botulinum toxin, augmentation or vesicostomy if needed.
- **Caveat:** 40 is not a validated "safe" value; 20 cmH_2O is more sensitive in children.

Also know: DO leak point pressure (ICS (International Continence Society) 2017); MUCP (maximum urethral closure pressure) ≤20 cmH_2O suggests ISD.

Diagram to draw: Pves, Pabd and Pdet traces marking ALPP and DLPP.

Key points to remember:

- ALPP low = weak sphincter; DLPP high = upper-tract danger.
- VLPP <60 ISD; ≥90 hypermobility.
- DLPP >40: 85% upper-tract change (McGuire 1981).
- Lower DLPP is safer; no validated safe cut-off.

Q3f. Role of renal nephrometry in management of RCC.

Type: SAQ (10 marks)

Definition

- **Nephrometry scores** grade the anatomical complexity of a renal mass on CT (computed tomography) or MRI (magnetic resonance imaging), to standardise RCC (renal cell carcinoma) reporting and plan treatment.

R.E.N.A.L. score (Kutikov and Uzzo, 2009)

Component	1 point	2 points	3 points
Radius (maximum diameter)	≤4 cm	>4 to <7 cm	≥7 cm
Exophytic/endophytic	≥50% exophytic	<50% exophytic	Entirely endophytic
Nearness to collecting system or sinus	≥7 mm	>4 to <7 mm	≤4 mm
Anterior/posterior	Suffix a, p or x (no points)		
Location to polar lines	Entirely above or below	Crosses a polar line	>50% across, crosses midline, or entirely between

- Suffix "h": touches main renal vessels. **Total 4–12:** low 4–6, moderate 7–9, high 10–12.

PADUA (Preoperative Aspects and Dimensions Used for an Anatomical classification) score (Ficarra, 2009)

- Polar location, exophytic rate, renal rim (lateral or medial), renal sinus, collecting system and size, plus anterior/posterior face. **Total 6–14:** 6–7 low, 8–9 intermediate, ≥10 high risk.

C-index (centrality index)

- Distance from tumour centre to kidney centre ÷ tumour radius. **Lower value = more central, more complex.**

Role in management

Decision	Use
Partial vs radical nephrectomy	Feasibility and risk of PN (partial nephrectomy) vs RN (radical nephrectomy)
Approach	Open vs minimally invasive; hilar clamping
Predict outcomes	Warm ischaemia, blood loss, complications, conversion, collecting-system entry; urine leak (R and N components)
Function	Parenchymal loss, eGFR (estimated glomerular filtration rate) decline
Ablation	Cryoablation complications rise with complexity
Pathology	Higher scores: malignancy, higher grade

- **AUA (American Urological Association) 2021:** with high oncological potential, prefer RN only if PN would be challenging even for experts, there is no CKD (chronic kidney disease) or proteinuria, and new baseline eGFR will likely exceed 45.

Limitations

- Interobserver variation; ignores patient factors and surgeon experience.

Also know: ABC (arterial based complexity), DAP (diameter–axial–polar) scores; **MAP (Mayo Adhesive Probability) score** for adherent perinephric fat.

Diagram to draw: kidney with polar lines and the five R.E.N.A.L. components.

> **Key points to remember:**
>
> - RENAL 4–12: low 4–6, moderate 7–9, high 10–12.
> - PADUA 6–14; C-index lower = more central.
> - Predicts PN feasibility, ischaemia time and complications.
> - MAP score: adherent perinephric fat.

Rapid Revision Sheet – 2020-Summer Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe clinical implication of AIDS on the Genito-Urinary Tract? (LAQ)

- AIDS = CD4 <200; UTI risk rises as CD4 falls.
- HIV-TB: disseminated, less fibrosis, fewer strictures.
- Indinavir stones: radiolucent, may be CT-invisible; stop drug and hydrate.
- PEP within 72 hours for 28 days; circumcision cuts acquisition 50–60%.

Q2. Describe embryological development of VU Junction and outline management of primary VUR? (LAQ)

- Low (early) bud → lateral orifice → VUR plus dysplasia; high (late) bud → ectopia.
- Tunnel length to diameter 5:1 (Paquin).
- RIVUR: CAP halves recurrent UTI, not scarring.
- Open reimplantation 95.9%; Cohen over 98%.

Q3a. Berger's Disease. (SAQ)

- Commonest primary GN; haematuria with upper respiratory infection.
- Normal complement; biopsy shows mesangial IgA.
- Oxford MEST-C; 20–30% ESKD.
- ACE inhibitor or ARB is the mainstay.

Q3b. Describe instrumentation for development of extraperitoneal space in laparoscopy? (SAQ)

- Space is created first: finger, then balloon.
- Gaur balloon: glove on red rubber catheter.
- Balloon trocar seals and prevents emphysema.
- Balloon in muscle causes hernia.

Q3c. Lichen sclerosis. (SAQ)

- Genital skin disease; meatus is the sentinel site.
- Spares proximal urethra; perineal urethrostomy easy.
- Clobetasol first; oral mucosa, never skin, for repair.
- SCC in 2.3–9%: lifelong follow-up.

Q3d. Renal allograft rejection. (SAQ)

- Hyperacute rejection prevented by crossmatch.
- Banff AMR = injury + C4d + DSA.
- TCMR: pulse steroids, ATG; AMR: plasmapheresis, IVIG.
- Chronic active AMR is the main cause of late loss.

Q3e. Leak point pressures-clinical implication. (SAQ)

- ALPP low = weak sphincter; DLPP high = upper-tract danger.
- VLPP <60 ISD; ≥90 hypermobility.
- DLPP >40: 85% upper-tract change (McGuire 1981).
- Lower DLPP is safer; no validated safe cut-off.

Q3f. Role of renal nephrometry in management of RCC. (SAQ)

- RENAL 4–12: low 4–6, moderate 7–9, high 10–12.
- PADUA 6–14; C-index lower = more central.
- Predicts PN feasibility, ischaemia time and complications.

- MAP score: adherent perinephric fat.